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1) Write a PL/SQL block that uses a cursor attribute SQL%ROWCOUNT to raise the basic salary of employees by 10% that are working in department number 10 and also display the appropriate message based on the existence to the record in the EMP table. (Use Implicit Cursor)

```
SET SERVEROUTPUT ON;

DECLARE
BEGIN
    UPDATE emp
    SET sal = sal + (sal * 0.10)
    WHERE deptno = 10;

    IF SQL%ROWCOUNT > 0 THEN
        DBMS_OUTPUT.PUT_LINE(SQL%ROWCOUNT || ' employees salary updated in department 10');
    ELSE
        DBMS_OUTPUT.PUT_LINE('No employees found in department 10');
    END IF;
END;
/
```

2) Write a PL/SQL block that uses a cursor attribute %ISOPEN and %NOTFOUND to raise the basic salary of employees of department number 20 by 5% and also display the appropriate message based on the existence to the record in the EMP table. Whenever any such raise is given to the employees, a record for the same is maintained in the emp_update table. (Perform using both Implicit and Explicit Cursor)

--Using Implicit Cursor

```
SET SERVEROUTPUT ON;

DECLARE
BEGIN
    UPDATE emp
    SET sal = sal + (sal * 0.05)
    WHERE deptno = 20;

    IF SQL%ROWCOUNT > 0 THEN
        INSERT INTO emp_update
        SELECT empno, sal, SYSDATE
        FROM emp
        WHERE deptno = 20;

        DBMS_OUTPUT.PUT_LINE('Salary updated for employees in department 20');
    ELSE
        DBMS_OUTPUT.PUT_LINE('No employees found in department 20');
    END IF;
END;
/
```

2) Write a PL/SQL block that uses a cursor attribute %ISOPEN and

%NOTFOUND to raise the basic salary of employees of department number 20 by 5% and also display the appropriate message based on the existence to the record in the EMP table. Whenever any such raise is given to the employees, a record for the same is maintained in the emp_update table. (Perform using both Implicit and Explicit Cursor)

--Using Explicit Cursor

```
SET SERVEROUTPUT ON;

DECLARE

    CURSOR c1 IS
        SELECT empno, sal
        FROM emp
        WHERE deptno = 20;

    v_empno emp.empno%TYPE;
    v_sal    emp.sal%TYPE;
BEGIN
    OPEN c1;

    IF c1%ISOPEN THEN
        LOOP
            FETCH c1 INTO v_empno, v_sal;
            EXIT WHEN c1%NOTFOUND;

            UPDATE emp
            SET sal = sal + (sal * 0.05)
            WHERE empno = v_empno;

            INSERT INTO emp_update
            VALUES (v_empno, v_sal, SYSDATE);
        END LOOP;

        DBMS_OUTPUT.PUT_LINE('Salary updated and record inserted into emp_update table');
    END IF;

    CLOSE c1;
END;
/
```

3) Write a PL/SQL block that uses a cursor attribute %ROWCOUNT to display the name, department and basic salary of first 5 employees getting the highest basic salary. (Use Explicit Cursor)

```
SET SERVEROUTPUT ON;

DECLARE

    CURSOR c1 IS
        SELECT ename, deptno, sal
        FROM emp
        ORDER BY sal DESC;

    v_name    emp.ename%TYPE;
    v_dept    emp.deptno%TYPE;
    v_sal     emp.sal%TYPE;
BEGIN
    OPEN c1;

    LOOP
        FETCH c1 INTO v_name, v_dept, v_sal;
        EXIT WHEN c1%NOTFOUND OR c1%ROWCOUNT > 5;

        DBMS_OUTPUT.PUT_LINE('Name: ' || v_name || ' Department: ' || v_dept || ' Salary: ' ||
v_sal);
    END LOOP;

    CLOSE c1;
END;
/
```

4) Write a PL/SQL block using a cursor FOR loop to display name and the basic salary of top 3 highest paid employees. (Use Cursor For Loop)

```
SET SERVEROUTPUT ON;

DECLARE
    CURSOR c1 IS
        SELECT ename, sal
        FROM emp
        ORDER BY sal DESC;

    counter NUMBER := 0; BEGIN

        FOR rec IN c1 LOOP
            counter := counter + 1;
            EXIT WHEN counter > 3;

            DBMS_OUTPUT.PUT_LINE('Name: ' || rec.ename ||
                                ' Salary: ' || rec.sal);

        END LOOP;
    END;
/
```

5) Write a PL/SQL block using a parameterized cursor that displays the department wise basic salary of each employee and department wise total gross salary. (Parameterized Cursor and Use Cursor For Loop)

```
SET SERVEROUTPUT ON;

DECLARE
    CURSOR emp_cursor(p_deptno NUMBER) IS
        SELECT ename, sal
        FROM emp
        WHERE deptno = p_deptno;

    total_sal NUMBER;
BEGIN
    FOR d IN (SELECT DISTINCT deptno FROM emp) LOOP
        total_sal := 0;          DBMS_OUTPUT.PUT_LINE('Department:
' || d.deptno);

        FOR e IN emp_cursor(d.deptno) LOOP
            DBMS_OUTPUT.PUT_LINE('Employee: ' || e.ename || ' Salary: ' || e.sal);
            total_sal := total_sal + e.sal;
        END LOOP;

        DBMS_OUTPUT.PUT_LINE('Total Salary of Department ' || d.deptno || ' = ' || total_sal);
        DBMS_OUTPUT.PUT_LINE('-----');
    END LOOP;
END;
/
```